# Experiment -2

Study of different types of network cables and crimping of cable with RJ45 connector

Aim:

To study different types of network cables and to prepare a straight-through and cross-over Ethernet cable by crimping RJ45 connectors

Requirements:

UTP cat5e / cat6 cable, RJ45 connectors, crimping tool, cable stripper, cable tester, scissors

Cable types:

1) UTP → unshield twisted pair (UTP)
   - category 3 - 10 mbps
   - category 5 - upto 100 mbps
   - category 5e - upto 1gb ps

2) STP → shield twisted pair (STP) cable
   - category 6 - upto 10gbps
   - SSTP - category 7 - upto 10 gbps

3) coaxial cable

   RG-6, RG-59, RG-11

max data transmission 10-100 mbps

crimping of network cable with RJ45 connector

RJ45 -

| | |
|---|---|
| 1 | wire |
| 2 | white green |
| 3 | green |
| 4 | white orange |
| 5 | blue |
| 6 | white blue |
| 7 | orange |
| 8 | Brown |

4) fibre optics cable:

single mode multimode

max data transmission - 100 Gbps

Procedure:

i) To start construction of the device, begin by threading

ii) next, strip approximately 1.5cm of cable shielding from both ends. the crimping tools has a rounded area to complete this task

iii) After, you will need to untangle the 8 four twisted pairs.

iv) once the order is concepts, bunch them together in line

v) To test it plug it in and attempt to connect two devices directly

Result:

Thus, the types of network cables are deided and crimped the cable with RJ45 connector is successfully written